V.S.B. ENGINEERING COLLEGE

(An Autonomous Institution)

KARUDAYAMPALAYAM, KARUR – 639 111.

RECORD NOTE BOOK

Name : _____

Reg. No. :_____

Subject : _____

Semester : _____

V.S.B. ENGINEERING COLLEGE

(An Autonomous Institution)

KARUDAYAMPALAYAM, KARUR – 639 111.

_____ RECORD

NAME:_____CLASS: _____

REGISTER NO. _____

ROLL NO. _____

Certified that this is a bonafide record of work done by the above student during the year 20 - 20

Date: Lab in-charge Head of the Department

Submitted for the "END SEMESTER PRACTICAL EXAMINATION"

held on …….

Internal Examiner External Examiner

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V.S.B. Engineering College, Karur
(An Autonomous Institution)
Department of Computer Science and Business Systems

VISION, MISSION, POs, PEOs and PSOs

Vision of the Institution:

We endeavour to impart futuristic technical education of the highest quality to the student community andto inculcate discipline in them to face the world with self-confidence and thus we prepare them for life as responsible citizens to uphold human values and to be of service at large. We strive to bring of the Institution as an Institution of academic excellence of International standard.

Mission of the Institution:

We transform persons into personalities by the state-of the art infrastructure, time consciousness, quick response and the best academic practices through assessment and advice.

VISION OF THE DEPARTMENT:

To produce professionals as industry ready in the field of computer science and business system with strong human values to serve the nation.

MISSION OF THE DEPARTMENT:

The Department strives to contribute to the expansion of knowledge in the discipline of Computer Science and Business System by,

- Offer a well-designed and updated curriculum that covers essential principles and practical skills in information technology. This should include technical knowledge, programming expertise, database management, network administration, cybersecurity, and emerging technologies. Ensure a balance between theoretical concepts and hands-on experience through practical assignments, projects, and internships.
- Provide opportunities for students to develop their soft skills, including effective communication, teamwork, leadership, and emotional intelligence. Offer workshops and extracurricular activities that promote self-confidence, creativity, and problem-solving abilities.
- Encourage students and faculty members to engage in research and innovation in the field of information technology. Provide resources and support for research projects, collaborations, and participation in national/international conferences. This will foster a culture of continuous learning, creativity, and technological advancement.
- Embed the importance of ethical values, integrity, and social responsibility in the institution's culture. Encourage community service, social initiatives, and sustainable practices that contribute to the betterment of society.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates can

PEO #1: Choose a specific engineering discipline or field based on personal interest and market demand. Specializing in areas such as mechanical engineering, electrical engineering, civil engineering, chemical engineering, or computer engineering will allow for in-depth knowledge and expertise.

PEO#2: Embrace a mindset of continuous improvement. Seek feedback, learn from failures, and strive for excellence in all aspects of engineering work. Continuously upgrade technical expertise, professional skills, and knowledge to stay ahead in a competitive industry.

PEO #3: Develop strong problem-solving skills, a critical attribute for engineers. Enhance analytical thinking, logical reasoning, and systematic approaches to problem-solving through practice, case studies, and projects. This helps in formulating effective and efficient solutions for industrial problems.

PROGRAM SPECIFIC OUTCOMES (PSOs)

The Students will be able to

PSO#1: Based on the identified requirements and process analysis, design BPM solutions that address the specific needs of the industry or social sector. This involves defining the desired state of the processes, documenting process flows, creating system specifications, and designing user interfaces.

PSO#2: Develop the BPM solutions based on the defined design and system specifications. This may involve custom development, configuration of BPM software, or integration with existing systems. Ensure a systematic approach to development, including thorough testing, debugging, and quality assurance.

PSO#3: Foster collaboration within the industry or social sector to share best practices and lessons learned in BPM. Encourage collaboration among organizations, academia, and relevant stakeholders to collectively seek innovative solutions and address common challenges.

V.S.B. ENGINEERING COLLEGE
KARUR - 639 111
HARDWORK IS THE KEY TO SUCCESS

Program Outcomes (PO)

PO 1: Engineering knowledge: Apply the Mathematical knowledge and the basics of Science and Engineering to solve the problems pertaining to Computer Science and Engineering.

PO2: Problem analysis: Identify and formulate Computer Science and Engineering problems from research literature and be able to analyze the problem using first principles of Mathematics and Engineering Sciences

PO3: Design/development of solutions: Come out with solutions for the complex problems and to design system components or process that fulfil the particular needs taking into account public health and safetyand the social, cultural and environmental issues.

PO4: Conduct investigations of complex problems: Draw well-founded conclusions applying the knowledge acquired from research and research methods including design of experiments, analysis and interpretation of data and synthesis of information and to arrive at significant conclusion.

PO5: Modern tool usage: Form, select and apply relevant techniques, resources and Engineering and ITtools for Engineering activities like electronic prototyping, modelling and control of systems and also being conscious of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Computer Science and Engineering practice.

PO7: Environment and sustainability: Be aware of the impact of professional Engineering solutions in societal and environmental contexts and exhibit the knowledge and the need for sustainable Development.**PO8: Ethics:** Apply the principles of Professional Ethics to adhere to the norms of the engineering practice and to discharge ethical responsibilities.

PO9: Individual and team work: Function actively and efficiently as an individual or a member/leaderof different teams and multidisciplinary projects.

PO10: Communication: Communicate efficiently the engineering facts with a wide range of engineering community and others, to understand and prepare reports and design documents; to make effective presentations and to frame and follow instructions

PO11: Project management and finance: Demonstrate the acquisition of the body of engineering knowledge and insight and Management Principles and to apply them as member / leader in teams and multidisciplinary environments.

PO12: Life-long learning: Recognize the need for self and life-long learning, keeping pace with technological challenges in the broadest sense

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS
CCW332-DIGITAL MARKETING LABORATORY

COURSE OUTCOMES (COs)

After completion of the course, Students are able to,

COs	Knowledge Level	Course Outcomes
1.	K4	Examine and explore the role and importance of digital Marketing in today's rapidly changing business environment.
2.	K3	Focuses on how digital marketing can be utilized by Organizations and how its effectiveness can be measured
3.	K2	Know the key elements of a digital marketing strategy
4.	K2	Study how the effectiveness of a digital marketing campaign Can be measured
5.	K4	Demonstrate advanced practical skills in common digital Marketing tools such as SEO, SEM, Social media and Blogs

List of Experiments Mapping with COs, POs &PSOs

Exp. No	Name of the Experiment	COs	POs	PSOs
1.	Subscribe to a weekly/quarterly newsletter and analyse how its content and structure aid with the branding of the company and how it aids its potential customer	CO1	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO2
2.	Perform keyword search for a skincare hospital website based on search volume and competition using Google keyword planner tool.	CO2	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO2

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

Exp. No	Name of the Experiment	COs	POs	PSOs
3.	Demonstrate how to use the Google WebMasters Indexing API	CO2&CO3	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO1, PSO2,PSO3
4.	Discuss an interesting case study regarding how an insurance company manages leads.	CO3	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO1, PSO2,PSO3
5.	Discuss negative and positive impacts and ethical implications of using social media for political advertising.	CO4 & CO5	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO1, PSO2,PSO3
6.	Discuss how Predictive analytics is impacting marketing automation	CO5	PO1, PO2,PO3, PO4, PO5, PO10 PO11, PO12	PSO2

Open Ended Experiments

Exp.No	Name of the Experiment	COs	Pos	PSOs
1.	Digital Marketing Implementation in Business Scenario	CO1 & CO2	PO6,PO7,PO8,PO9	PSO1, PSO2,PSO3
2.	Create the Digital Marketing Webpage	CO1	PO6,PO7,PO8,PO9	PSO3
3.	Facebook Promotion using banners	CO3 & CO4	PO6,PO7,PO8,PO9	PSO1, PSO3
4.	Twitter Marketing	CO4	PO6,PO7,PO8,PO9	PSO1, PSO2,PSO3
5.	Instagram Marketing	CO4	PO6,PO7,PO8,PO9	PSO1, PSO2,PSO3

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

Program Outcomes:

1.	**Engineering Knowledge**	**2.**	**Problem Analysis**
3.	**Design/Development of Solutions**	**4.**	**Conduct Investigations of Complex Problems**
5.	**Modern Tool Usage**	**6.**	**The Engineer and Society**
7.	**Environment and Sustainability**	**8.**	**Ethics**
9.	**Individual and Team Work**	**10.**	**Communication**
11.	**Project Management and Finance**	**12.**	**Life-long Learning**

Program Specific Outcomes:

PSO#1: Addressing societal problems through design and development of software and firmware solutions using latest Computer Science tools and technologies.

PSO#2: Involving enthusiastically in software development, software testing, storage, computing and business intelligence sectors.

PSO#3: Use their technical expertise in latest technologies and update their knowledge continuously in Computer Science and Engineering to excel in their career.

Ex.No:1

Date:

Subscribe to a weekly/quarterly newsletter and analyze how its content and structure aid with the branding of company and how its potential customer segments.

Aim:

To Subscribe to a weekly/quarterly newsletter and analyze how its content and structure aid with the branding of company and how its potential customer segments.

Introduction:

This study focuses on the analysis of a company's weekly/quarterly newsletter to understand how its content and structure contribute to branding efforts and cater to specific customer segments. By examining the newsletter's strategies, we can uncover how it effectively engages subscribers and aligns with the company's branding goals.

Objective:

To analyze a company's weekly/quarterly newsletter to understand how its content, structure, and segmentation strategies contribute to branding efforts and cater to specific customer segments.

Materials:

- Access to the company's newsletters
- Notebook or digital document for analysis
- Survey tool for subscriber feedback

Procedures :

Branding Alignment:

- Review the Newsletter's Tone and Voice: Analyze whether the tone and voice of the newsletter align with the company's brand identity. Is it professional, casual, formal, or friendly?
- Consistency with Brand Elements: Look for consistent use of brand colors, logos, fonts, and other visual elements throughout the newsletter.
- Brand Messaging: Evaluate if the messaging in the newsletter reflects the company's core values, mission, and unique selling propositions.

Content Strategy:

- Relevance of Topics: Examine if the content of the newsletter is relevant to the company's products, services, and industry. Does it provide valuable insights to subscribers?

- Customer-Centric Approach: Identify whether the newsletter addresses customer pain points, offers solutions, and provides useful information.
- Storytelling: Assess whether the newsletter uses storytelling techniques to engage readers and connect them emotionally with the brand.

Customer Segments:

- Target Audience: Analyze who the primary target audience of the newsletter is. Does it cater to existing customers, potential leads, or a specific demographic?
- Personalization: Look for personalized content, such as addressing subscribers by their names or tailoring content to specific customer segments.
- Call-to-Action (CTA): Observe if the newsletter includes CTAs that resonate with different customer segments, encouraging them to take specific actions.

Frequency and Consistency:

- Newsletter Schedule: Evaluate the frequency of the newsletter (weekly, quarterly, etc.) and whether it aligns with the preferences of the target audience.
- Consistent Delivery: Check if the newsletter is consistently delivered on time, helping subscribers anticipate its arrival.

Engagement and Interactivity:

- Analyze if the newsletter encourages engagement, such as asking for feedback, conducting surveys, or promoting social media interaction.
- Interactive Elements: Look for interactive elements like polls, quizzes, or clickable content that increase reader engagement.

Value-Added Content:

- Identify if the newsletter offers exclusive content, such as special offers, discounts, or access to premium resources for subscribers.

Visual Appeal:

- Examine the use of images, graphics, and layout. Is it visually appealing and easy to read?

Conversion Focus:

- Assess whether the newsletter guides subscribers towards taking desired actions, such as visiting the website, making a purchase, or signing up for webinars.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Result:

Through this lab experiment, we gained insights into how a company's newsletter content, structure, and segmentation strategies contribute to its branding goals and engage specific customer segments. The experiment underscores the importance of personalized content, consistent branding, and strategic CTAs in building a strong brand image and connecting with the target audience.

The above program is mapped with CO1, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, PSO1,PSO2 and PSO3.

Ex.No: 2

Date:

Perform keyword research for a skincare hospital website based on search volume and competition using Google keyword Planner

Aim:

To Perform keyword research for a skincare hospital website based on search volume and competition using Google keyword Planner

Introduction :

Keyword research is a process of finding words, queries, and phrases that users are searching for, which means a keyword that has search volume. Research involves connecting the relevance of keywords to a website and its individual pages so that the user can find the best page to answer their query, known as search intent. Keyword research also involves categorizing search queries into the different stages of a user journey and different categories of search, such as transactional, navigational, and informational.

Good keyword research enables users to find what they need:

- Shoppers who want to buy something can find the right product page.
- A user that wants to know 'how to' can find a page that explains a process in-depth.
- Users who want to research a person or brand can find out about that entity.

Procedures :

Access Google Keyword Planner: Log in to your Google Ads account, or create one if you don't have it already. Then, go to the Google Keyword Planner tool.

Start a New Search: Click on "Discover new keywords" or "Get search volume and forecasts."

Enter Seed Keywords: Enter the seed keywords that are relevant to your skincare hospital. For example: "skincare hospital," "dermatology clinic," "skin treatments," etc.

Filter and Customize:

a) Specify your target location (city, region, country) to get localized results.
b) Adjust the date range and language preferences.

View Keyword Ideas: Google Keyword Planner will provide you with a list of keyword ideas related to the seed keywords you entered. It will show you metrics like average monthly searches and competition level.

Analyze Metrics:

a) Look for keywords with a good balance of search volume and competition.

b) Focus on keywords that are relevant to your services and have a substantial search volume.
c) Consider long-tail keywords (more specific phrases) that might have lower competition.

Competition Level: Google Keyword Planner provides a competition level indicator (low, medium, high). Aim for keywords with medium to low competition.

Ad Group Ideas: Google Keyword Planner will also suggest ad group ideas. These can provide insight into how Google categorizes keywords.

Negative Keywords: Consider adding negative keywords to filter out irrelevant terms that might not be suitable for your skincare hospital.

Export Data: Export the selected keywords along with their metrics to a spreadsheet for further analysis.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Result:

Conducting keyword research using Google Keyword Planner is a crucial step in optimizing a skincare hospital website for search engines. By identifying keywords with relevant search volume and manageable competition, the skincare hospital can strategically target its content to attract potential patients seeking specialized skincare services.

The above program is mapped with CO2, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, PSO1,PSO2 and PSO3.

Ex.No: 3

Date:

Demonstrate how to use Google Webmasters indexing API

Aim:

To demonstrate how to use Google Webmasters indexing API.

Introduction :

The Google Webmasters Indexing API is a powerful tool that allows you to directly notify Google about changes to your website's content for quicker indexing. Below, I'll provide you with an outline of the steps to demonstrate how to use the Google Webmasters Indexing API. To demonstrate the usage of the Google Webmasters Indexing API to notify Google about changes to a website's content for quicker indexing.

Materials:

- Google Cloud Platform account with API Key
- Access to Google Search Console
- cURL (command-line tool for making HTTP requests)

Procedure :

Steps to Use Google Webmasters Indexing API:

Enable the Indexing API:

- Go to the Google Cloud Console (https://console.cloud.google.com/).
- Create a new project if you don't have one.
- Enable the "Indexing API" for your project.

Generate API Key:

- In the Cloud Console, navigate to "APIs & Services" > "Credentials."
- Create a new API Key.

Add Permissions to API Key:

- Configure the API Key to have access to the "Indexing API" by selecting the API under "API restrictions."

Integrate API Key:

- In your code, use the API Key to authenticate your requests to the Indexing API.

Send an Indexing Request:

- Construct a JSON payload with the necessary information. You'll need the URL of the updated/added page and optionally a timestamp.
- Make a POST request to the Indexing API endpoint with the constructed payload

Sample request using cURL:

```
curl -X POST -H "Content-Type: application/json" \
-H "Authorization: Bearer YOUR_API_KEY" \
-d '{
  "url": "https://www.example.com/new-page",
  "type": "URL_UPDATED"
}' \
"https://indexing.googleapis.com/v3/urlNotifications:publish"
```

Handle API Response:

- The API will respond with success or error messages. Handle the responses in your code.

Monitor Indexing Status:

- Check your Google Search Console to monitor the indexing status and progress of the submitted URLs.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Results and Discussion:

By following the outlined procedure, we successfully utilized the Google Webmasters Indexing API to notify Google about changes to our website's content. This approach can expedite the indexing process, ensuring that our updated or newly added pages are quickly included in Google's search index.

The Google Webmasters Indexing API provides a valuable tool for webmasters to directly communicate changes to their website's content to Google. By integrating this API into our workflow, we can enhance the efficiency of our indexing process, leading to more accurate and timely search results.

The above program is mapped with CO2, CO3, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, PSO1,PSO2 and PSO3.

Ex.No: 4

Date:

Discuss an interesting case study regarding how an insurance company manages leads

Aim:

To discuss an interesting case study regarding how an insurance company manages leads.

Introduction:

This case study delves into how an insurance company effectively manages leads through a strategic digital marketing approach. By leveraging digital channels, personalized communication, and data-driven strategies, the company achieved significant improvements in lead quality and conversion rates.

Challenge:

The insurance industry is highly competitive, with companies vying for potential clients' attention. The company faced challenges in efficiently identifying, nurturing, and converting leads within a crowded market.

Solution: The insurance company adopted a comprehensive digital marketing strategy that encompassed lead generation, segmentation, nurturing, and conversion optimization.

Key Strategies:

Targeted Lead Generation:

- The company utilized targeted online advertising campaigns on platforms like Google Ads and social media to reach potential customers with specific demographics and interests.
- Landing pages were optimized for lead capture, providing valuable content in exchange for contact information.

Data-Driven Segmentation:

- Leads were segmented based on factors such as age, location, income level, and insurance needs.
- This segmentation allowed for more personalized communication tailored to the specific needs of each lead.

Personalized Nurturing:

- Automated email workflows were set up to deliver relevant and educational content to leads over time.

- The content addressed common concerns, explained insurance concepts, and showcased the company's expertise.

Retargeting Campaigns:

- Leads who interacted with initial ads but didn't convert were retargeted with follow-up ads to keep the brand top-of-mind.

Social Proof and Testimonials:

- The company showcased success stories, customer testimonials, and positive reviews to build trust and credibility.

Results:

Increased Lead Quality:

- The digital marketing strategies attracted leads that were more aligned with the company's offerings, resulting in higher-quality leads.

Enhanced Conversion Rates:

- By delivering personalized content and maintaining consistent communication, the company achieved improved lead-to-customer conversion rates.

Reduced Acquisition Costs:

- Targeted advertising and personalized nurturing reduced acquisition costs by avoiding spending on uninterested leads.

Stronger Customer Relationships:

- The personalized approach led to stronger customer relationships, resulting in higher customer retention and referrals.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Conclusion:

Through a strategic blend of targeted lead generation, data-driven segmentation, personalized nurturing, and retargeting efforts, the insurance company effectively managed leads in the digital landscape. By delivering valuable content and building trust, the company not only increased conversion rates but also fostered lasting customer relationships.

The above program is mapped with CO3, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, PSO1,PSO2 and PSO3.

Ex.No: 5

Date:

Discuss the negative and positive impact and ethical implications of using social media for political advertising

Aim:

To discuss the negative and positive impact and ethical implications of using social media for political advertising.

Introduction:

This discussion explores the dual-sided impact of using social media for political advertising. It delves into the positive and negative consequences, as well as the ethical implications, of leveraging digital platforms for political messaging.

Positive Impact:

Reach and Accessibility:

- Social media allows political campaigns to reach a broader and more diverse audience, including segments that might not engage with traditional media.

Engagement and Interaction:

- Interactive features like comments, likes, shares, and polls enable direct engagement with voters, fostering two-way communication.

Cost-Effectiveness:

- Political advertising on social media can be more cost-effective compared to traditional media, making it accessible to smaller campaigns.

Microtargeting:

- Social media platforms offer advanced targeting options, enabling campaigns to reach specific demographics and tailor messages accordingly.

Negative Impact:

Spread of Misinformation:

- False information can spread rapidly on social media, leading to the dissemination of misleading or outright false political claims.

Polarization and Echo Chambers:

- Algorithms can contribute to echo chambers, reinforcing users' existing beliefs and exacerbating political polarization.

Privacy Concerns:

- The collection of user data for targeting purposes raises privacy concerns and ethical questions about the use of personal information.

Manipulation and Fake Accounts:

- Bad actors can manipulate social media through fake accounts, bots, and automated engagement, distorting public discourse.

Ethical Implications:

Transparency:

- Political advertisers should be transparent about the source, funding, and intent of their ads to maintain public trust.

Accountability:

- Campaigns must take responsibility for the accuracy and truthfulness of their content, avoiding intentionally deceptive tactics.

Influence and Manipulation:

- Manipulating emotions and exploiting psychological vulnerabilities through ad targeting raises ethical concerns about the potential for manipulation.

Impact on Democracy:

- The influence of social media on political opinions and elections raises questions about the health of democratic processes.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Result:

The use of social media for political advertising comes with both positive and negative impacts, alongside significant ethical implications. While it enhances reach, engagement, and targeting capabilities, it also poses risks related to misinformation, polarization, privacy, and manipulation. As digital marketers, it's crucial to consider the ethical dimensions of political advertising and strive for transparency, accuracy, and responsible engagement with the public.

The above program is mapped with CO4, CO5, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, PSO1,PSO2 and PSO3.

Ex.No: 6
Date:

Discuss how predictive analytics is impacting marketing automation

Aim:
To discuss how predictive analytics is impacting marketing automation.

Introduction:

Predictive analytics has revolutionized the field of marketing automation by enabling businesses to make data-driven decisions, enhance customer targeting, and optimize marketing campaigns. This discussion explores how predictive analytics is reshaping marketing automation strategies and outcomes.

Objective:

To investigate the impact of integrating predictive analytics into marketing automation on campaign effectiveness and customer engagement.

Materials:

- Marketing automation platform with predictive analytics capabilities
- Historical customer data
- Test campaigns with and without predictive analytics integration
- Analytics tools for data collection and analysis

Procedure:

Data Preparation:

- Gather historical customer data, including purchase history, interactions, and demographics.

Campaign Design:

- Design two test campaigns: one with predictive analytics integration and one without.
- Both campaigns should target the same customer segment with similar content.

Execution:

- Launch the campaigns through the marketing automation platform.
- The predictive analytics-integrated campaign should leverage insights for personalized timing and content.

Engagement Measurement:

- Monitor metrics such as open rates, click-through rates, and conversion rates for both campaigns.

Data Analysis:

- Analyze the engagement metrics to compare the effectiveness of the campaigns with and without predictive analytics integration.

Predictive Analytics in Marketing Automation:

Enhanced Customer Segmentation:

- Predictive analytics analyzes historical data to identify patterns and behaviors, enabling marketers to segment customers more effectively.
- Segmentation based on predictive insights allows personalized and relevant communication.

Behavioral Analysis and Lead Scoring:

- Predictive analytics examines customer behavior to assign lead scores based on the likelihood of conversion.
- Marketing automation systems prioritize high-scoring leads for tailored nurturing and engagement.

Content Personalization:

- Predictive analytics helps marketers understand customer preferences and interests.
- Marketing automation platforms leverage this insight to deliver personalized content, increasing engagement.

Churn Prediction:

- Predictive analytics forecasts which customers are at risk of churning.
- Marketing automation tools can trigger retention campaigns for identified high-risk customers.

Optimized Campaigns:

- Predictive analytics identifies the best times, channels, and content for specific customers.
- Marketing automation automates the execution of these optimized campaigns.

Particulars	Max Mark	Mark Obtained
Performance		
Viva Voce		
Record		
Total		

Result:

Through this lab experiment, we observed the tangible impact of integrating predictive analytics into marketing automation. The experiment highlights how predictive insights contribute to personalized targeting, optimized campaigns, and improved customer engagement. Emphasizing the symbiotic relationship between predictive analytics and marketing automation underscores the potential for data-driven marketing success.

The above program is mapped with CO5, PO1, PO2,PO3, PO4, PO5, PO10, PO11, PO12, and PSO2.

Open Ended Experiments

Experiment No. 1

Digital Marketing Implementation in Business Scenario

"Digital Marketing is the marketing of products or services using digital technologies on the Internet, including through mobile phones Apps, using display advertising, and any other digital mediums"

--- Damian Ryan and Calvin Jones

What is Digital Marketing mainly used for? Although SEO (Search Engine Optimization), PPC (Pay Per Click) and Web based Marketing still accounts for a major chunk of activities, other areas like CRO (Conversion Rate Optimization), Video Content Creation, App based marketing and Blog based marketing are fast catching up.

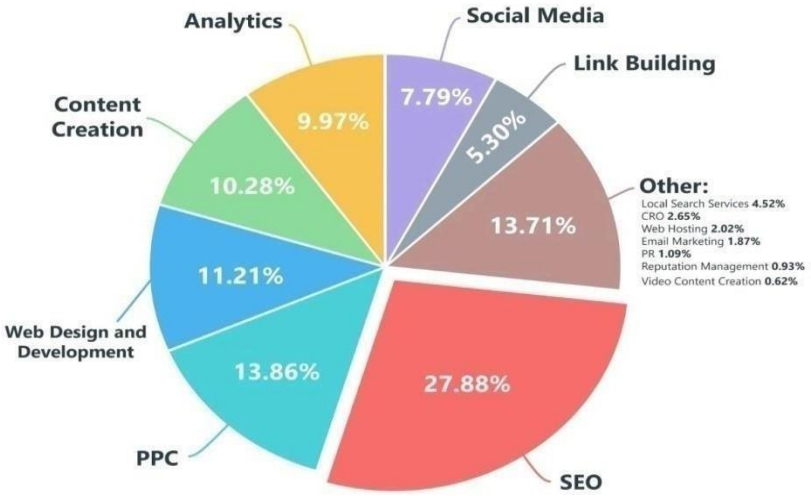
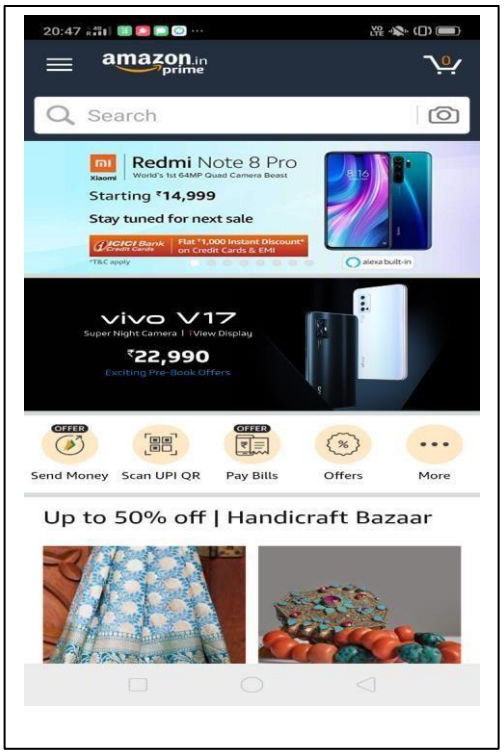
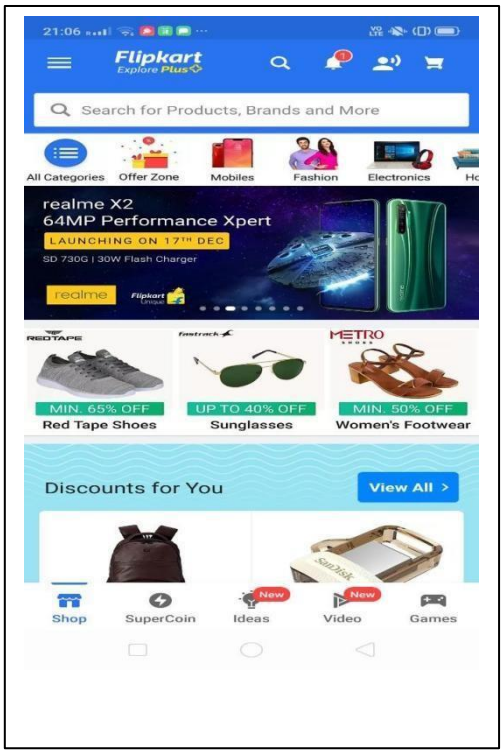


Figure- 1: Use of Digital Marketing Tools

It is now well accepted that Digital Marketing has become a key component of business strategy for all organizations, big or small. Digital Marketing is playing a key role in E-Commerce Companies as well as brick-and-mortar companies. However, the challenge now is to know the correct strategies that will drive maximum returns with minimum investments. The first phase of Digital Marketing has seen companies adopting strategies like SEO (Search Engine Optimization), SEM (Search Engine marketing), SMM (Social Media Marketing) and E-Mail Marketing. The main platforms being used were Google, Facebook, YouTube and Twitter.

The students are expected to perform the following tasks for understanding the strategies used by companies for digital marketing:





1. Open the websites of Amazon, Flipkart, Snapdeal
2. Do a comparative analysis of their product offerings
3. Do a comparative analysis of their landing pages
4. Do a comparative analysis of their call to action (CTA)
5. Do a comparative analysis of website loading and website navigation
6. Find the rankings of Amazon, Flipkart, Snapdeal using Alexa.com

Result:

Through this lab experiment, we observed the Digital Marketing Implementation in Business Scenari by various online companies and how they promote their products.

Experiment No. 2
Create the Digital Marketing Webpage

For Digital Marketing or E-Commerce, a Website is the virtual store from where the commercial transaction of buying and selling takes place. The Website gives information about the products and services; the price tag associated with various products and services and provides a mechanism to acquire the products and services through electronic exchange of money. A webstore created for E-Commerce needs to have the following components:

1. Home Page that contains information about products and services
2. About Page that contains detailed description about the company
3. Contact Us Page where the Contact Number, Fax Number, Email, Twitter and Facebook contact details all are given.
4. Payment Gateway to make payments through Debit Card or Credit Cards.
5. SSL (Secure Socket Layer) Certificate. SSL Certificates provide secure, encrypted communications between a website and an internet browser.

Like people differ from each other, a webstore layout can also differ according to the need of target users and creativity of the web designer. However, a content heavy website with rich graphics will take a long time in loading. A simple website might not be able to catch the attention. This where the rules of optimization becomes important to create a website that is pleasing to the eye and also loads faster than the average ones.

The students are expected to perform the following tasks to create the promotional web page

1. Go to GoDaddy and analyze the different kind of domain names offered there.

2. Go to GoDaddy and analyze the different kind of hosting options offered there.

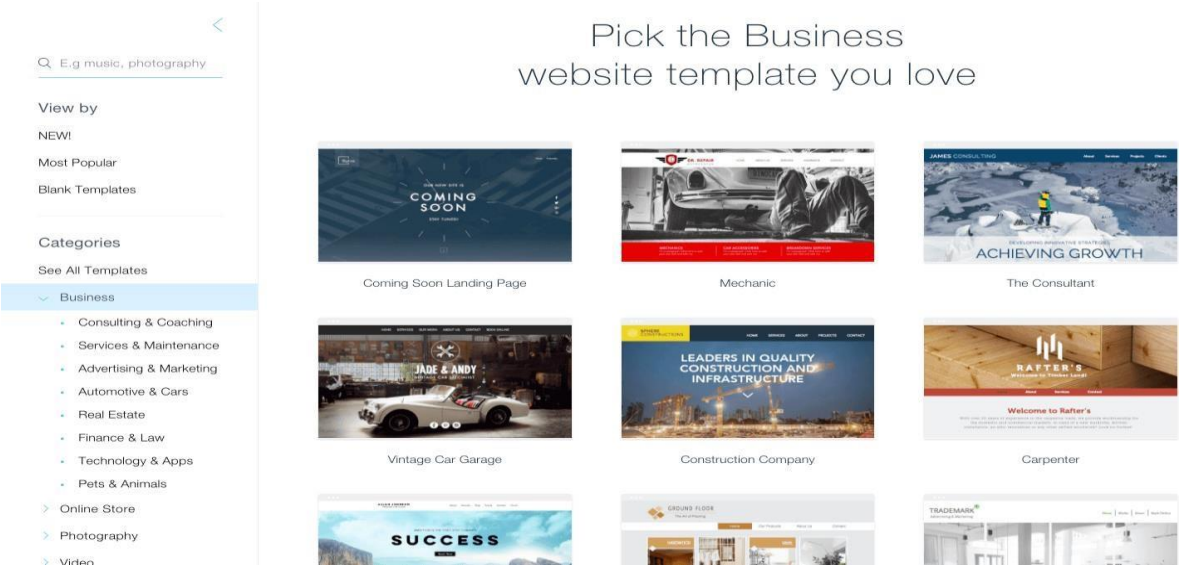
3. Go to Wix.com and create a promotional web page in a shared hosting service

Wix.com is a website builder that allows you to build a website with no coding. Wix is free to start but to set up a shop, remove branding, connect a custom domain, and more, you must select one of the affordable premium plans.

4. Customize the Website using various options given by Wix

5. Publish the website online

6. Find the ranking of your website using Alexa.com



Result:

Through this lab experiment, we know about how to Create the Digital Marketing Webpage.

Experiment 3
Facebook Promotion using banners

Facebook is a social networking website that allows registered users to create profiles, upload photos and video, send messages and keep in touch with friends, family and colleagues. The main features of Facebook are News Feed and Facebook Messenger. News Feed is the constantly updating list of stories in the middle of your home page. News Feed includes status updates, photos, videos, links, app activity and likes from people, pages and groups that you follow on Facebook. News Feed also shows advertisements in the middle of the pages.

Facebook Messenger is an instant messaging service and software application. Facebook Messenger lets Facebook users send messages to each other. Complementing regular conversations, Messenger lets users make voice calls and video calls both in one-to-one interactions and in group conversations.

A Facebook Page is a public profile specifically created for businesses, brands, celebrities, causes, and other organizations. Unlike personal profiles, pages do not gain "friends," but "fans" - which are people who choose to "like" a page. Pages can gain an unlimited number of fans, differing from personal profiles, which has had a 5,000 friend maximum put on it by Facebook. Pages work similarly to profiles, updating users with things such as statuses, links, events, photos and videos. This information appears on the page itself, as well as in its fans' personal news feeds.

Facebook Ad is an advertisement created by a business on Facebook that’s served up to Facebook users based on user activity, demographic information, device use information, advertising and marketing partner-supplied information, and off-Facebook activity. Facebook Ad can be created by Ad Manager. You can choose the audience and goal and allocate a budget. The Ad can be in the form of an image, canvas, video, slideshow, collection or carousel. The Ad can be displayed in Facebook Page, Instagram, Messenger and Facebook Audience Network.

Facebook Groups are the place for small group communication and for people to share their common interests and express their opinion. Groups allow people to come together around a common cause, issue or activity to organize, express objectives, discuss issues, post photos and share related content. When you create a group, you can decide whether to make it publicly available for anyone to join, require administrator approval for members to join or keep it private and by invitation only. Like with Pages, new posts by a group are included in the News Feeds of its members and members can interact and share with one another from the group.

A Facebook Event is a calendar-based resource which can be used to notify users of upcoming occasions. Events can be created by anyone, and can be open to anyone or private. The creator can invite his friends, members of a group, or fans of a page. Facebook events are a great way to spread the word on upcoming events or occasions, since they are able to reach thousands of people in a short amount of time. The event also provides an "RSVP" list, which displays lists of invitees grouped by their response.

Invitees are either placed in "attending," "not attending," "may be attending," or "hasn't responded" lists. If an invitee RSVPs that they are "attending" the event, it appears in their news feed to notify their friends. When the date of the event approaches, it is displayed on the invitees' home pages to remind them.

Tasks to be done

1. Go to facebook.com/pages/create.

Click to choose a Page type.

Fill out the required information.

2. Choose a great username.

3. Give the key details in the About section.

4. Provide as many details as possible in the full About description.

5. Capture attention with a fantastic cover photo visual.

6. Choose an appropriate profile photo.

7. Ask your Facebook friends to like your Facebook Page.

Once your Page is setup, use the handy invite option to invite all of your Facebook friends (or a group of your Facebook friends) to like your Facebook Page via a Facebook direct message. This is a great way to get the ball rolling and start building a following.8.Promote your Facebook Page.

8. Link your Facebook Page with your Webpage in Wix.com

9. Use Facebook Insights to analyze the impact of your facebook campaign

Result:

Through this lab experiment, we know about how to promote facebook using banners.

Experiment 4

Twitter Marketing

1. Create a Twitter Profile

Your Twitter profile shows the world who you are, and first impression counts. Each element should showcase your best content and accurately reflect your message. To create a Twitter profile, you need to choose @Name, put up your profile picture, write a description with 160 characters, and put up a nice attractive header image.

There are certain rules and specifications that you need to comply with to create a Twitter profile. The recommended image size for your profile picture should be 400x400 pixels; the image will be resized in mobile to fit. The recommended size for a Twitter header image is 1500x500 pixels. On mobile, it'll be cropped to a 2:1 aspect ratio. You can use a GIF or PNG file for vector-based and line art images. For photos, you can upload a JPG or PNG file.

2. Create compelling content to Tweet

Compelling content will help you attract new followers and keep them engaged over time. There's no right or wrong number of times you should Tweet each day, or when. Instead, focus on creating a regular flow of content that's relevant to your target audience and authentic to your business.

Adding a bold image, video, or GIF to your Tweets adds a touch of personality, and leads to higher Tweet engagement rates. In fact, people are three times more likely to engage with Tweets that contain videos and photos. Twitter allows you to attach up to four photos to a single Tweet.

3. Create a Tweeting Strategy/Schedule

Similar to planning for digital marketing, you should also have a set posting schedule for your Twitter account. Your Tweeting schedule should detail what and when you plan on Tweeting. Your strategy should outline things like when you're going to Tweet links to your new posts and other updates.

It's a good idea to include this as a part of your overall content strategy, particularly if you publish content across several different platforms like a blog, YouTube, guest posts and media coverage. A benefit of having a set Tweeting strategy is you can avoid mistakes like only Tweeting out links to your new blog posts once.

4. Make Tweets Conversational

The way that a lot of brands tweet is very one dimensional. Their Tweets are essentially just broadcasts, which is not what Twitter is about as we mentioned earlier. Your Tweets shouldn't consist of only headlines with a link, inspirational quotes or funny statements.

They should open the door for communication and conversation.

5. Use Twitter Video

Twitter might not be the first platform you think of when it comes to video marketing, but it's not something you should overlook. Twitter gives you a couple of options for using videos.

First, you can use Twitter's native video feature. This allows you to record videos up to 140 seconds long and upload them directly to your Twitter stream.

A second option is to use Periscope, which is a live streaming app owned by Twitter. Periscope integrates into the Twitter so your live streams show up in your followers' Twitter feeds. Once the stream is over, the recording is still available for people to watch.

Live video from Twitter's content partners reached over 31 million views in 2017, so it's safe to say your followers are likely interested in consuming your videos.

Connect with existing and potential customers

People love to talk to businesses directly on Twitter. The brand manager should be responsive to any questions, comments, and criticisms that come through tweets. To avoid long exchanges, switch to Direct Messages to resolve any complex issues. You can also include a deep link in a Tweet that displays a 'Send a private message' call-to-action button to enable customers to send you a Direct Message.

Hashtags are a powerful tool that allows you to expand your reach and tap into relevant conversations. Focus on keywords that are relevant to your business. Best practices recommend using no more than two hashtags per Tweet. One simple way to incorporate hashtags is by identifying popular events to which you can link your business in an authentic way.

Retweeting relevant content and replying to Tweets are great ways to maintain a robust Twitter presence. Positive customer feedback, helpful articles, and messages that align with your business's authentic voice are all impactful content to Retweet. When in doubt, remember this rule of thumb: your Retweets reflect back on your business and should align with your purpose and values.

Result:

Through this lab experiment, we know about how to to do marketing in twitter using banners.

Experiment 5

Instagram Marketing

Instagram was created by Kevin Systrom and Mike Krieger, and launched in October 2010 as a free mobile app for photo editing and photo sharing exclusively for the iOS operating system. A version for Android devices was released two years later, in April 2012. After its launch in 2010, Instagram rapidly gained popularity, with one million registered users in two months, 10 million in a year, and ultimately 800 million as of September 2017. Today, over five million businesses use Instagram to tell their stories visually, connect with their fans, and build their brand Instagram was acquired by Facebook in April 2012 for approximately $1 billion in cash and stock.

Instagram allows users to upload photographs and short videos, follow other users' feeds, and geotag images with the name of a location. In January 2011, Instagram introduced hashtags to help users discover both photos and each other. In September 2011, a new version of the app included new and live filters, instant tilt–shift, high-resolution photographs, optional borders, one-click rotation, and an updated icon. In August 2015, Instagram started allowing users to upload full-size landscape and portrait photos and videos onto the service, dropping the previous requirement of a square frame. In August 2016, Instagram added a zoom feature that allows users to pinch-to-zoom the screen to virtually zoom in on photos and videos.

Tasks in Instagram Marketing

1. Create an Instagram Account
2. Post the Banner Ad in Instagram
3. Use filters to make the banner more attractive
4. Write compelling captions
5. Use the right hashtags
6. Run the Instagram Campaign
7. Measure the success with Instagram Analytics

Result:

Through this lab experiment, we know about how to to do marketing in Instagam..